Ram Operator Banks for Triple Ergodic Averages with Distinct Integer Slopes

Source-Specific Reformulation, Quantitative Planting, and Conditional Pointwise Extensions

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Abstract

We analyze the pinned manuscript on triple ergodic averages at arbitrary distinct nonzero integer slopes, including mixed signs. We state its depth-scheduled local estimate with all uniformity requirements and map the coefficient algebra, quadratic localization, rational-shift counting, absorption, and orbit transference on which its claim depends. The Ram representation is an exact bank of Koopman iterates followed by physical products and scale contraction, not a replacement for multilinear analysis. For the actual planting region we compute its area and all even step moments from the extreme slopes. A two-radius Richardson combination then gives an explicit fourth-order summed discretization error, at the cost of a larger leading transfer constant. We prove signed-time finite-horizon perturbation bounds and the conditional passage from the local estimate to general-system convergence, weak-mixing product limits, and bounded affine or periodic variants. Deterministic block endpoints are retained throughout. Classical results and the prior private reports are attributed; no global analytic closure, effective mixing rate, historical novelty, or computational speedup is claimed. Contribution assessment: C2/5, explicit local planting and signed-slope stability refinements are proved, but the universal multilinear estimate remains a separate verification obligation.

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1 Problem, quantifiers, and evidence status

Let $(X, \mathcal{F}, \mu)$ be a probability space and let T be an invertible, bimeasurable, measure-preserving map. On the complex Hilbert space $H = L^2(\mu)$ use $Uf = f \circ T$ and $\langle f, g\rangle = \int f\overline{g}\, d\mu$. Then U is unitary, has domain all of H, and U^m exists for every $m \in \mathbb{Z}$. On every L^p, including L^∞, these same powers are isometries. Multiplication is used on bounded functions, not as an everywhere-defined operation $H \times H \to H$.

For fixed $\lambda = (\lambda_1, \lambda_2, \lambda_3) \in \mathbb{Z}^3$ with $\lambda_i \neq 0$ and $\lambda_i \neq \lambda_j$ for $i \neq j$, put

$$A_N^\lambda(\mathbf{f})(x) = \frac{1}{N}\sum_{n=1}^{N}\prod_{j=1}^{3} U^{\lambda_j n} f_j(x), \qquad N \in \mathbb{Z}_{>0}. \tag{1.1}$$

The source [1] claims that for *every* such system satisfying

$$\mu(A \cap T^{-m}B) \longrightarrow \mu(A)\mu(B) \quad (|m| \to \infty), \qquad A, B \in \mathcal{F}, \tag{1.2}$$

and *every* bounded measurable triple, A_N^λ tends almost everywhere, through all positive integers, to $\prod_j \int f_j\, d\mu$. There is no quantitative mixing hypothesis, no spatial boundary condition, and no standardness assumption. The null set may depend on both the functions and the slopes. Countably many prescribed triples and slopes may share a conull set by intersection; an uncountable universal exceptional-set assertion is not being made.

Normalize $\|f_j\|_\infty \leq 1$ when proving estimates. The zero-norm case is trivial; otherwise multiply bounds by $\prod_j \|f_j\|_\infty$. If functions are equivalence classes, select bounded representatives and discard the countable union of integer translates of the exceptional sets. Define

$$g = \gcd(|\lambda_1|, |\lambda_2|, |\lambda_3|), \quad S = T^g, \quad b_0 = 0, \quad b_j = \lambda_j/g. \tag{1.3}$$

This is an exact reduction, including negative slopes. All later analytic constants may depend on this fixed b; no uniform bound over integer triples is asserted. Replacing T by a power and permuting inputs does not convert general b to $(1, 2, 3)$.

1.1 Three levels of assertion

The source is a new public preprint, pinned at

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

This report distinguishes: (i) what that preprint claims; (ii) elementary identities and local estimates proved here; and (iii) consequences conditional on its universal local estimate. Reading the source and checking coefficient identities is not independent verification of the entire imported analytic machinery. The completed independent source audit gives a bounded pass to the deduction of deterministic-block oscillation from the local theorem, including signed and nonunimodular slopes. It does not give an unconditional pass to that theorem: the necessary H/V dependency chain was not fully audited. It found no counterexample to the source claim. Incomplete certification is an audit limitation, not a demonstrated mathematical gap. Section 9 records the review's actual scope separately from the author's proofs.

The catalogue's **L4** is a relevance assessment. The present **C2** is a contribution assessment: a little contribution, not a significant advance relative to the full pinned claim. Neither rating certifies validity or historical priority. The prior M295 report is used as a comparison and source of ideas, with its selected arguments checked again where used, not as an automatic theorem oracle.

1.2 Current primary literature check

The public check was made on October 8, 2026, using only public titles, identifiers, and URLs. No private passage was sent in a query. Walsh's Theorem 1.1 [6] gives L^2 convergence for bounded polynomial averages from a nilpotent group; taking the cyclic group generated by T and polynomials $\lambda_j n$ includes both signs here. This is norm convergence, not a pointwise estimate.

Bergelson's weak-mixing norm result [7] applies to distinct nonconstant linear polynomials. Its mechanism is also proved below. The publisher record was checked, but its full theorem text was not available in that interface. The actual theorem and introduction of Kosz–Mirek–Peluse–Wan–Wright, version 4 of September 2, 2026 [8], were read: their pointwise theorem assumes pairwise *distinct degrees*. The three polynomials $\lambda_j n$ all have degree one. Using T^{λ_j} as commuting transformations does not change those degrees. Thus that theorem cannot supply the missing estimate for this source.

Krause–Mirek–Tao [9] concerns a bilinear average with a polynomial of degree at least two; its publisher statement was checked. Gutman, Huang, Shao, and Ye [10], whose Main Theorem was read in the primary PDF, prove the progression product limit under weak mixing *and* the pairwise-independent-joining determination property. We do not silently remove that property or generalize their displayed progression coefficients. These checks establish boundaries of the cited results, not an exhaustive literature search or confirmation that the new OpenAI preprint has been independently accepted. The attempted Frantzikinakis progress-page read failed in the browser; no current claim rests on that unsuccessful request.

2 The precise local analytic premise

Write $y_j = x + b_j t$. A finite translated dyadic tree $\mathcal{T}$ contains all ancestors between its cells and its root J. Its largest depth is d and $s(I) = \log_2(|J|/|I|)$. For $r = |I|$, fix $\delta_0 > 0$ and $C_0 \geq 1$, and require $w_I \in C_c^\infty(\mathbb{R}^2; \mathbb{C})$ to satisfy

$$y_j \in I, \quad \operatorname{dist}(y_j, \partial I) \geq \delta_0 r \qquad ((x,t) \in \operatorname{supp} w_I,\ 0 \leq j \leq 3), \tag{2.1}$$

$$\int_{\mathbb{R}} w_I(v - b_j t, t)\, dt = 0 \qquad (v \in \mathbb{R},\ 0 \leq j \leq 3), \tag{2.2}$$

$$\|\partial_x^a \partial_t^d w_I\|_\infty \leq r^{-a-d}[C_0(a+d+2)]^{C_0(a+d+2)} \qquad (a, d \in \mathbb{Z}_{\geq 0}). \tag{2.3}$$

The reuse of d as a derivative index in (2.3) has no relation to tree depth. The scalar four-linear form is

$$H_I(z_0, z_1, z_2, z_3) = r^{-2} \int_{\mathbb{R}^2} w_I(x,t) \prod_{j=0}^{3} z_j(y_j)\, dx\, dt. \tag{2.4}$$

It is defined whenever the integral is absolutely convergent, in particular for the bounded zero-extended schedules below.

For an integer block $B = [u, v] \cap \mathbb{Z}$ set

$$\mathcal{V}_B(z) = \max_{u \leq s \leq v} |z_s| + \sum_{s=u}^{v-1} |z_{s+1} - z_s|. \tag{2.5}$$

Each slot j has its own deterministic partition of $\{0, \ldots, d\}$ into $W_j \geq 1$ nonempty consecutive blocks. The values $z_{j,s} : J \to \mathbb{C}$ may depend measurably on position, but the endpoints do not. At every original depth, including unused depths, require $\mathcal{V}_B(z_j) \leq W_j^{-1/2}$ almost everywhere, for every block. Extend the functions by zero outside J.

Definition 2.1 (Premise L_b)**.** For every tree, subcollection $\mathcal{I} \subset \mathcal{T}$, admissible kernel assignment, and schedules just specified,

$$\sum_{I \in \mathcal{I}} |I|\, |H_I(z_{0,s(I)}, z_{1,s(I)}, z_{2,s(I)}, z_{3,s(I)})| \le C_{b,\delta_0,C_0} |J|. \tag{2.6}$$

The constant is independent of the translated lattice, tree, depth, kernel choices within the fixed class, partitions, and all W_j.

This is exactly the mathematical scope of the source's `thm:local`, not a proved new theorem here. It has no mixing hypothesis. An estimate for fixed inputs, a single finite tree, or a fixed number of changes is weaker than L_b.

2.1 A source proof map

All locators in this subsection are the literal labels in the pinned source, not labels in this report. The eight files were read as mathematics, not only as section inventories.

1. `main.tex` and `sections/introduction.tex`: the headline `thm:main`, signed slopes, arbitrary probability spaces, and the explicit warning that the companions' $(0, 1, 2, 3)$ forms do not prove the arbitrary-coefficient statement.
2. `sections/local.tex`: normalization, all kernel hypotheses, `thm:local`, whole-configuration integer reconstruction `lem:algebra`, and the fixed-input constants $C_N(q)$ for $2 < q < 3$.
3. `sections/transfer.tex`: `lem:tr-kernel` realizes scale differences by four directional derivatives; `lem:tr-real` averages translated lattices; `lem:tr-integer` plants sequences using the integer left inverse; `prop:tr-oscillation` uses deterministic blocks and spatially selected prefixes; `lem:tr-cauchy` gives convergence; `lem:tr-norm` identifies the mixing limit; `lem:tr-density` and the final proof recover every N by signed profiles and rational grids.
4. `sections/analytic.tex`: `lem:local-absolute` uses pair determinants. Chart separation and detection are `lem:progression-separation` and `lem:local-detection`. Normalized curvature tags use the Vandermonde weights. The separation and same-tag tests feed inherited projections, main/helper catalogs, `eq:analytic-closure`, and localized group expressions. `eq:analytic-endpoint-variation` is a schedule-uniform row claim. The exact band-transfer identity `eq:analytic-band-transfer` retains the band sum inside each cell; separate absolute values can destroy the stated gain.
5. `sections/arithmetic.tex`: `prop:short-block-b` imports fixed-input control with cost $(1 + C_N(q))C_{q,b}(1+\log(K+2))^C(K+2)^{11(q-2)/(q-1)}$. Fast Fourier randomization uses $\sum k_j^2/c_j = 0$. The common gauge and rational shift lattice lead to `lem:height-b`, with a base D_{s} divisible by 6 and every pair difference. `eq:near-band-b` has the crucial $P^{-1/2}$ gain and strong overhead G_0, not the large mode budget D_*. `lem:sparse-b` asserts a sparse-mass power $1/2080$, loss L^{29}, and reserves polynomial in dimension. The five hypotheses include pair determination, side incidence, separation, triple annuli, and a pure-shift gap; a Vandermonde identity alone supplies none of these counting conclusions. The final core estimate is `eq:core-final-b`.
6. `sections/completion.tex`: retained-cell estimates, restarts with total root length at most $4|R_0|/3$, and the main/mode budget hierarchy precede the choice of q. Fixed-input absorption precedes the scheduled-input proof; it is not legitimate to use the final scheduled theorem to establish its own fixed-input constant.
7. `references.tex`: distinguishes classical transference, norm and pointwise literature, the inverse theorem, and the two new companion manuscripts H and V.

The quantitative closure being claimed is worth retaining explicitly:

$$\text{retained cost}/|R| \le C_q\eta^{\varepsilon}P^{C_4} + (1+C_N(q))C_qP^{-1/2}G_0(K+2)^{C_{19}(q-2)} + O(e^{-k}), \tag{2.7}$$
$$\eta = e^{-k^{\theta}}, \quad 0<\theta<1, \quad K \le C_q(1+k+P)^{C_5}, \quad G_0 \le e^{C_q(1+\log(2k))^C}.$$

The degrees must be fixed before taking q sufficiently near 2 that $C_{19}C_5(q-2) < 1/8$. Then $P = \lceil \eta^{-\alpha} \rceil$ with $\alpha C_4 < \varepsilon/2$ gives a claimed positive quality power. The level series has terms $te^{-c_* k_t^{\theta}}$ with $k_{t+1} = k_t^A$, $A > 1$. After normalization the source obtains $C_N(q) \le CA_\tau + C\tau C_N(q)$, and uses prior finiteness $C_N(q) \lesssim_b N$ to absorb when $C\tau < 1/2$. We record these dependencies to prevent a local algebra check from being mistaken for verification of (2.7).

3 Coefficient algebra and the Ram representation

3.1 The full integer configuration

Lemma 3.1 (Reconstruction and cancellation, after the source)**.** *For primitive b there are $e_j \in \mathbb{Z}$ and $c_j \in \mathbb{Z}\setminus\{0\}$ such that*

$$\sum e_j = 0, \quad \sum e_j b_j = 1, \qquad \sum c_j b_j^a = 0 \quad (a = 0,1,2). \tag{3.1}$$

Thus $x = y_0$, $t = \sum e_j y_j$, and $\sum c_j y_j^2 = 0$. The nullspace of the two rows $(1)_j, (b_j)_j$ is spanned by $(c_j)_j, (b_j c_j)_j$.

Proof. Use Bezout on b_1, b_2, b_3 and set $e_0 = -\sum_{j=1}^3 e_j$. For $P(z) = \prod_j (z - b_j)$, Lagrange interpolation of z^a, $a \le 2$, and comparison of the cubic coefficient give $\sum_j b_j^a / P'(b_j) = 0$. Clear denominators to obtain c. The two null vectors are independent since all c_j are nonzero and the b_j are distinct. The matrix has rank two. Substituting $y_j = x + b_j t$ proves the remaining assertions. □

The pair map has determinant $b_j - b_i$: it is injective on $\mathbb{Z}^2$, but may not map onto $\mathbb{Z}^2$; on the real torus it is onto, but may not be one-to-one. These are different properties. A concrete primitive nonunimodular example with integer reconstruction and cancellation is

$$b = (0,6,10,15), \quad e = (-1,1,1,-1), \quad c = (-6,25,-27,8). \tag{3.2}$$

They satisfy (3.1) directly. No pair difference is ± 1.

3.2 Banks, physical products, and Gram coordinates

The actual private TR-item-168 defines a Ram decomposition by structured, uniform, long-variation and short-variation blocks, and later a finite bank $f \mapsto (U^n f)_{n\in I}$ with physical products. We use the latter representation because it is exact for arbitrary signed slopes, without assuming a pointwise characteristic-factor theorem.

For a finite set $I \subset \mathbb{Z}$, define

$$\mathcal{R}_{\lambda,I} : (L^\infty(\mu))^3 \longrightarrow (L^\infty(\mu))^{3|I|}, \qquad \mathcal{R}_{\lambda,I}(\mathbf{f})_{j,n} = U^{\lambda_j n} f_j,$$
$$\mathcal{P}(v)_n = v_{1,n} v_{2,n} v_{3,n}, \qquad \mathcal{C}_a(w) = \sum_{n\in I} a_n w_n. \tag{3.3}$$

Here $a \in \mathbb{C}^I$. The first map is linear on the product space, the second is trilinear between its three slot banks, and the third is linear. Their composite is exactly the weighted average. Taking $I = [1,N]$ and $a_n = 1/N$ gives (1.1); smooth scales use $a_n = 2^{-k}\varphi(n/2^k)$ on their finite support.

Proposition 3.2 (Exact bounded-bank representation)**.** *On one-bounded input banks,*

$$\|\mathcal{C}_a\mathcal{P}(v)\|_\infty \le \sum |a_n|, \quad \|\mathcal{C}_a(\mathcal{P}(v)-\mathcal{P}(w))\|_2 \le \sum_n |a_n| \sum_{j=1}^{3} \|v_{j,n}-w_{j,n}\|_2. \tag{3.4}$$

If $v_{j,n} = \sum_{r=1}^{D} a_{j,n,r}\phi_r$ *in a declared finite physical model, then products are synthesized by* $\sum_{r,s,t} a_{1,n,r}a_{2,n,s}a_{3,n,t}\phi_r\phi_s\phi_t$*. They are not generally coordinatewise products of the vectors* $a_{j,n}$*.*

Proof. The first assertion is the triangle inequality. Expand a product difference as $(v_1-w_1)v_2v_3 + w_1(v_2-w_2)v_3 + w_1w_2(v_3-w_3)$, and use bounded multipliers on L^2, then the triangle inequality over n. Expanding the finite synthesis gives the last identity. □

For linearly independent $\phi_r \in L^\infty$, let $G_{rs} = \int \overline{\phi_r}\phi_s \, d\mu$. Synthesis Φa has physical energy a^*Ga, so a linear analysis matrix B on this model is bounded by C exactly when $B^*B \preceq C^2G$. Indeed the matrix inequality means $\|Ba\|^2 \le C^2\|\Phi a\|_2^2$ for every a. It says nothing about the operator on the model's orthogonal complement. This is elementary Gram geometry, consistent with the actual finite-model warnings in TR-item-168 and M295; it is not new harmonic analysis.

On $\mathbb{Z}/q\mathbb{Z}$, $T(x) = x+1$, Fourier characters give an exact model:

$$A_N^b(\mathbf{f})(x) = \sum_{r_1,r_2,r_3 \in \mathbb{Z}/q\mathbb{Z}} \widehat{f}_1(r_1)\widehat{f}_2(r_2)\widehat{f}_3(r_3)e_q((r_1+r_2+r_3)x)\frac{1}{N}\sum_{n=1}^{N} e_q(n\textstyle\sum_j b_j r_j). \tag{3.5}$$

This follows by Fourier inversion and finite summation. Resonances are $\sum b_j r_j = 0 \pmod q$. Finite Fourier diagonalization of the orbit bank does not diagonalize physical multiplication or yield a uniform in-q oscillation estimate. No FFT speedup is measured here.

4 Exact signed-slope planting and a fourth-order refinement

The refinement in this section is unconditional local analysis. It does not assume L_b until a real-line bound is explicitly transferred.

4.1 No spurious lattice configurations

Let $E_b = \sum_j |e_j|$, $B_b = \max_j |b_j|$, and choose

$$0 < \rho < 1/4, \qquad \rho(2 + B_bE_b) < 1. \tag{4.1}$$

Plant a sequence value at m on $[m-\rho, m+\rho)$ and put zero elsewhere. If $x + b_jt$ lies in planted intervals centered at integers p_j, write $\delta_j = x + b_jt - p_j$. Set $m = p_0$, $n = \sum e_jp_j$. Then $x = m+\xi$, $t = n+\eta$ with $\xi = \delta_0$ and $\eta = \sum e_j\delta_j$, and

$$|p_j - m - b_jn| \le |\delta_j| + |\xi| + |b_j\eta| < \rho(2 + B_bE_b) < 1.$$

The integer on the left is zero. Conversely the real configurations with these centers have offset region

$$P_\rho = \{(\xi,\eta) : |\xi + b_j\eta| < \rho \ (0 \le j \le 3)\}. \tag{4.2}$$

Boundaries have measure zero. Different integer center pairs give disjoint regions up to boundaries. This is the source's reconstruction argument, with its dependence on the whole primitive configuration retained.

Proposition 4.1 (Extreme-slope geometry)**.** *Put* $b_- = \min_j b_j$, $b_+ = \max_j b_j$, *and* $d_b = b_+ - b_- > 0$. *Then*

$$\begin{aligned} A_\rho := |P_\rho| &= \frac{4\rho^2}{d_b}, \quad |\eta| \le \frac{2\rho}{d_b}, \\ \frac{1}{A_\rho}\int_{P_\rho} \eta^{2m} &= \frac{2^{2m+1}\rho^{2m}}{(2m+1)(2m+2)d_b^{2m}} \quad (m \ge 0). \end{aligned} \tag{4.3}$$

and every odd step moment is zero. In particular the second and fourth moments are $2\rho^2/(3d_b^2)$ *and* $16\rho^4/(15d_b^4)$.

Proof. Every $\xi + b_j\eta$ is a convex combination of the two extreme coordinates. Thus all intermediate constraints are redundant, even when the slopes have both signs. The change $u = \xi + b_-\eta$, $v = \xi + b_+\eta$ maps P_ρ onto $(-\rho, \rho)^2$, has Jacobian d_b, and gives $\eta = (v-u)/d_b$. Under normalized area, u, v are independent uniform variables on that interval. Their difference has density $(2\rho - |s|)/(4\rho^2)$ on $[-2\rho, 2\rho]$. Hence the even moment is

$$\frac{2}{4\rho^2 d_b^{2m}} \int_0^{2\rho} s^{2m}(2\rho - s)\, ds,$$

which equals (4.3). Symmetry gives the odd moments. □

The $b = (0, 1, 2, 3)$ case and second-order Taylor idea occur in M295. Here the full extreme-slope formula treats the assigned coefficients; its derivation is an elementary change of variables, not a novelty claim.

For $f \in C^r(\mathbb{R})$ define the normalized planting operator

$$K_\rho f(t) = A_\rho^{-1} \int_{P_\rho} f(t + \eta)\, d\xi\, d\eta. \tag{4.4}$$

Its domain can also be bounded measurable functions, but derivative estimates below use bounded derivatives of the indicated order.

Theorem 4.2 (Second- and fourth-order planting)**.** *For* $f \in C^2$ *with bounded second derivative,*

$$\|K_\rho f - f\|_\infty \le \frac{\rho^2}{3d_b^2}\|f''\|_\infty. \tag{4.5}$$

For $f \in C^4$ *with bounded fourth derivative, let* $\mathcal{K}_\rho = (4K_{\rho/2} - K_\rho)/3$. *Then*

$$\|\mathcal{K}_\rho f - f\|_\infty \le \frac{\rho^4}{54d_b^4}\|f^{(4)}\|_\infty. \tag{4.6}$$

The latter operator reproduces polynomials through degree three exactly and has sup-norm operator bound at most $5/3$.

Proof. Taylor's formula to order one, vanishing mean step, and $\mathbb{E}\eta^2/2 = \rho^2/(3d_b^2)$ give (4.5). Expansion to order three gives $K_a f = f + a^2 f''/(3d_b^2) + R_a$, where $\|R_a\|_\infty \le (2a^4/(45d_b^4))\|f^{(4)}\|_\infty$. The second-derivative terms cancel in $4K_{\rho/2} - K_\rho$. The remainder coefficient is

$$\frac{1}{3}\left(4\frac{2(\rho/2)^4}{45d_b^4} + \frac{2\rho^4}{45d_b^4}\right) = \frac{\rho^4}{54d_b^4}.$$

The mass coefficient is one, odd moments vanish, and the quadratic moment cancels, proving exact cubic reproduction. Each K_a is a positive averaging operator, so the triangle inequality gives $5/3$. □

This is Richardson extrapolation applied to the actual planting operator. It is not a new general extrapolation principle. The output is signed; positivity is not retained.

4.2 The precise summed error and its cost

Let $\psi \in C_c^4(\mathbb{R})$ with $\operatorname{supp}\psi \subset [-B, B]$, $B \geq 1$, and $\Delta_k(t) = 2^{-k}\psi(2^{-k}t)$, $k \geq 0$. Let $a_k(m,n) = h_{0,k}(m)\prod_{j=1}^{3} h_j(m + b_j n)$, with $|a_k(m,n)| \leq M_0$, and at most L possible spatial centers m. Write $D_k = \sum_{m,n} a_k(m,n)\Delta_k(n)$; let $I_k(a)$ be the real-line integral of the corresponding radius-a planted inputs against Δ_k. Both $a = \rho, \rho/2$ obey (4.1). Then

$$I_k(a)/A_a = \sum_{m,n} a_k(m,n) K_a \Delta_k(n). \tag{4.7}$$

Define $J_k = (4I_k(\rho/2)/A_{\rho/2} - I_k(\rho)/A_\rho)/3$ and

$$\begin{aligned} C_B &= 2B + 4\rho/d_b + 1, \\ E_2 &= \frac{\rho^2 C_B}{3d_b^2}\|\psi''\|_\infty, \qquad\qquad E_4 = \frac{\rho^4 C_B}{54 d_b^4}\|\psi^{(4)}\|_\infty. \end{aligned} \tag{4.8}$$

Corollary 4.3 (Summed planting errors)**.** *For every such finite input family and every $k \geq 0$,*

$$|I_k(\rho)/A_\rho - D_k| \leq LM_0E_2 2^{-2k}, \qquad |J_k - D_k| \leq LM_0E_4 2^{-4k}. \tag{4.9}$$

Consequently the tails from k_- cost at most $(4/3)LM_0E_2 2^{-2k_-}$ and $(16/15)LM_0E_4 2^{-4k_-}$, respectively.

Proof. Formula (4.7) follows from the disjoint regions. The rth derivative of Δ_k is bounded by $2^{-(r+1)k}\|\psi^{(r)}\|_\infty$. Only $|n| \leq B2^k + 2\rho/d_b$ contribute; their number is at most $C_B 2^k$. Apply Theorem 4.2 to each n, sum over m, n, and sum the two geometric series. This explains the difference between pointwise kernel orders $2^{-3k}, 2^{-5k}$ and summed orders $2^{-2k}, 2^{-4k}$. □

For the following transfer comparison, additionally assume that all spatial centers m lie in one interval of L consecutive integers, and that the normalized inputs satisfy the scheduled hypotheses of Section 5. Their radius-a plantings then have a common spatial support interval of length at most $L - 1 + 2a < L$. Cardinality alone would not imply this span bound; it is sufficient only for Corollary 4.3. Suppose, conditionally, the source real-line test gives $|\sum_k I_k(a)| \leq C_R(L + 2R)$ for both radii, with $R = D2^{k_+}$. This is valid under L_b with the same C_R, since planting preserves the pointwise schedule bounds and gives real support length at most L. The fourth-order discrete bound is then

$$\left|\sum_{k=k_-}^{k_+} D_k\right| \leq \frac{17C_R}{3A_\rho}(L + 2R) + \frac{16}{15}LM_0E_4\, 2^{-4k_-}. \tag{4.10}$$

Indeed $A_{\rho/2} = A_\rho/4$, so the leading coefficient is $(4 \cdot 4 + 1)/(3A_\rho) = 17/(3A_\rho)$. Single-radius second-order planting instead has leading coefficient C_R/A_ρ and its second-order tail. One may take the smaller bound. The faster tail does not automatically compensate for the leading constant, and it supplies no faster ergodic convergence rate.

5 From local cancellation to deterministic oscillation

This section reconstructs the assigned source's downstream argument. Its sole nonclassical global premise is L_b.

5.1 All four cancellations

A profile is a real $\varphi \in C_c^\infty(\mathbb{R})$ with mass one, moments 1 through 3 zero, and a derivative bound $\|\varphi^{(n)}\|_\infty \le [G(n+2)]^{G(n+2)}$ for some fixed G. Set $\varphi_a(t) = a^{-1}\varphi(t/a)$, $\psi = \varphi_2 - \varphi$, and $\Delta_k = \varphi_{2^{k+1}} - \varphi_{2^k}$. The moments of ψ through order three vanish. Its fourth primitive

$$q(t) = \frac{1}{6}\int_{-\infty}^{t} (t-v)^3 \psi(v)\, dv$$

is compactly supported: outside the right endpoint expand the cubic and use all four zero moments. It satisfies $q^{(4)} = \psi$. Take $u \in C_c^\infty([-1/8, 1/8])$ of mass one in the same derivative class, and a dyadic $D > 4BB_b$, where $\operatorname{supp}\psi \subset [-B, B]$. Define

$$W(z,\sigma) = \prod_{j=0}^{3} (\partial_\sigma - b_j\partial_z)[u(z)D^{-3}q(D\sigma)]. \tag{5.1}$$

Each factor is the derivative along $z = v - b_j\sigma$, so its line integral is zero. Support has $|z + b_j\sigma| < 3/8$; product rules give the required derivative class. Integrating in z leaves $D\psi(D\sigma)$. Thus on intervals of length $r = D2^k$, $w_I(x,t) = W((x-m_I)/r, t/r)$ is admissible and

$$r^{-1}\int_{\mathbb{R}} W(z, t/r)\, dz = \Delta_k(t). \tag{5.2}$$

For s_i the elementary symmetric functions of b_1, b_2, b_3, the actual differential polynomial in (5.1) is $\partial_\sigma^4 - s_1\partial_z\partial_\sigma^3 + s_2\partial_z^2\partial_\sigma^2 - s_3\partial_z^3\partial_\sigma$. This identity holds with negative coefficients and is tested exactly.

5.2 Real-line and integer tests

Fix $0 \le k_- \le k_+$ and a deterministic w-block partition. Let fixed $|h_j| \le 1$ and $h_{0,k}$ be supported on one real interval of length L, with $\mathcal{V}_B(h_0) \le w^{-1/2}$. Average translated dyadic lattices over a full period $R = D2^{k_+}$. At scale $r = D2^k$, unfolding the period gives

$$\frac{1}{R}\int_0^R \sum_{|I|=r} r^{-1} w_I(x,t)\, da = \Delta_k(t).$$

Only roots meeting the support contribute, with total length $\le L + 2R$. Depth is $k_+ - k$, so reversal preserves deterministic blocks and their variation. Applying L_b on those roots gives

$$\left|\sum_k \int h_{0,k}(x) \prod_{j=1}^{3} h_j(x + b_j t)\Delta_k(t)\, dx\, dt\right| \le C_R(L + 2R). \tag{5.3}$$

All sums involve finitely many scales and bounded supports. Plant integer sequences and apply (4.9) or (4.10). This yields an integer test

$$\left|\sum_{k,m,n} h_{0,k}(m) \prod_{j=1}^{3} h_j(m + b_j n)\Delta_k(n)\right| \le C_D(L + D2^{k_+}), \tag{5.4}$$

with C_D depending only on b and the fixed profile (and the local constant), not on blocks or endpoints.

Proposition 5.1 (Orbit oscillation, conditional on L_b)**.** *For any invertible probability-preserving S, any one-bounded triple, and $V_k = \sum_{n\in\mathbb{Z}} \varphi_{2^k}(n) \prod_j f_j(S^{b_j n}x)$, every deterministic $0 \le k_1 < \cdots < k_{w+1}$ satisfies*

$$\sum_{i=1}^{w} \int \max_{k_i \le \ell \le k_{i+1}} |V_\ell - V_{k_i}|\, d\mu \le 2C_D\sqrt{w}. \tag{5.5}$$

Proof. For finitely supported integer inputs form the analogous $V_k(m)$. At each spatial point choose the smallest maximizing $\ell_i(m)$ in the finite block, and let $\alpha_i(m)$ align the complex difference with the positive real axis (zero if the difference is zero). Set $h_{0,k}(m) = \alpha_i(m)/(2\sqrt{w})$ for $k_i \le k < \ell_i(m)$, and zero otherwise in that block. Its supremum and at most one internal jump sum to at most $w^{-1/2}$. Telescoping (5.4) gives the spatial oscillation sum bounded by $2C_D\sqrt{w}(L + D2^{k_{w+1}})$. The block endpoints are deterministic; only the prefix and phase depend on m. This factor 2 refinement of the source's factor 4 is already observed in M295.

Now fix the endpoints. If $\operatorname{supp}\varphi \subset [-B_\varphi, B_\varphi]$, choose an integer $Q \ge B_\varphi B_b 2^{k_{w+1}}$ and plant the orbit values $f_j(S^r x)$ for $1 - Q \le r \le M + Q$. Their support length is $M + 2Q$. For $1 \le m \le M$ no positive or negative iterate is lost. Integrate the spatial bound in x, use invariance, and divide by M. The right side is $2C_D\sqrt{w}[1 + (2Q + D2^{k_{w+1}})/M]$. Let $M \to \infty$ while the endpoint list and Q stay fixed. □

Lemma 5.2 (Deterministic tail-diameter criterion)**.** *Let Z_k be finite measurable complex functions. Suppose the left side of* (5.5)*, with Z replacing V, is bounded by $b(w) < \infty$ for every deterministic list. Then the extended tail diameter $\Omega = \lim_{K\to\infty} \sup_{p,q\ge K} |Z_p - Z_q|$ obeys*

$$\mathbb{E}\Omega \le 2 \inf_{w\ge 1} b(w)/w. \tag{5.6}$$

In particular $b(w) = C\sqrt{w} + \epsilon w$ implies $\mu\{\Omega > t\} \le 2\epsilon/t$; at $\epsilon = 0$ the sequence converges almost everywhere.

Proof. For each fixed K, $\sup_{\ell\ge K} |Z_\ell - Z_K| \ge \Omega/2$. For $0 < a < \mathbb{E}\Omega/2$, monotone convergence gives a deterministic $L > K$ with the finite maximum's expectation exceeding a. Repeat for w consecutive blocks to get $wa < b(w)$. Let a increase to $\mathbb{E}\Omega/2$. An infinite expectation contradicts finite $b(1)$ by the same argument. Markov's inequality gives the probability bound; zero tail diameter and finiteness imply the Cauchy property in $\mathbb{C}$. □

This refinement is inherited from M295's checked deterministic proof. It is included to specify the exact bridge used here, not counted as a new M294 result. No point-dependent crossing times are substituted for deterministic endpoints.

6 All integers and conditional extensions

6.1 Signed profiles and norm limits

The moment-correction argument in the source permits approximation of $h_c = c^{-1}\mathbf{1}_{(0,c]}$, $1 \le c \le 2$, in L^1 by profiles of the preceding section. Here is the full construction. Convolve h_c with a narrow, mass-one Gevrey bump ρ_δ to obtain v with mass one and $\|v - h_c\|_1 < \varepsilon/2$. Fix four distinct z_i and a mass-one bump ρ_0. The polynomials $P_j(z) = \int (z+u)^j \rho_0(u)\, du$ are monic of degree j. Thus $M = (P_j(z_i))_{j,i=0}^3$ is invertible by Vandermonde interpolation. Let $m_j = \int t^j v(t)\, dt$ and solve

$$M\beta = (0, -m_1/A, -m_2/A^2, -m_3/A^3)^T.$$

Then $\varphi = v + \sum_i \beta_i A^{-1}\rho_0(t/A - z_i)$ has the required exact moments and $\sum|\beta_i| = O_v(A^{-1})$. Choose A large. Finite dilation preserves the derivative class. This is an inherited source construction, not an additional result. There is no uniform support or derivative constant as $\varepsilon \to 0$. Nonnegativity is impossible: mass one and zero second moment for a smooth nonnegative function would force its support to be $\{0\}$.

For a general system put $F_n = \prod_j f_j(T^{\lambda_j n}x)$ and let L_+ and L_- be the classical L^2 limits of the positive and negative Cesaro averages, respectively. Both have modulus at most one, but they are not assumed equal. For any $\chi \in C_c^1(\mathbb{R})$, Abel summation separately on the two half-lines gives

$$\sum_{n\in\mathbb{Z}} a^{-1}\chi(n/a)F_n \longrightarrow \left(\int_0^\infty \chi\right) L_+ + \left(\int_{-\infty}^0 \chi\right) L_- \quad \text{in } L^2, \quad a \to \infty. \tag{6.1}$$

Indeed the partial sums of $F_{\pm n} - L_\pm$ have norm $o(n)$; summation by parts bounds their weighted error by $a^{-2}\|\chi'\|_\infty \sum_{n\le Ba+1} o(n) = o(1)$ for a fixed support bound B. The zero term is $O(a^{-1})$, and Riemann sums give the two masses. For a mass-one profile approximating h_c, its smooth limit L_φ therefore satisfies

$$\|L_\varphi - L_+\|_\infty \le \|\varphi - h_c\|_1(\|L_+\|_\infty + \|L_-\|_\infty).$$

This is another way to complete the profile argument, as noted in the parent review. The proof below instead uses norm convergence only after the positive skeleton has been shown Cauchy; both methods preserve the negative-time distinction.

Theorem 6.1 (Conditional convergence on arbitrary systems)**.** *Fix λ as in Section 1 and suppose L_b holds for its primitive b. Then every bounded triple on every invertible probability-preserving system has almost-everywhere convergent averages* (1.1)*. The limit is its classical L^2 limit and is not necessarily constant.*

Proof. Choose profiles $\varphi_{c,h}$ approximating h_c within 2^{-h} for rational $c \in [1,2]$ and integers $h \ge 1$. Proposition 5.1 and Lemma 5.2 give convergence of each smooth dyadic sequence. Intersect their countably many conull sets. For $N_k = \lfloor c2^k \rfloor$, direct comparison gives

$$|A_{N_k} - V_k^{\varphi_{c,h}}| \le (c2^k)^{-1} + 2^{-k}\sum_n |h_c(n/2^k) - \varphi_{c,h}(n/2^k)|. \tag{6.2}$$

The Riemann sum tends to the L^1 error, since the compactly supported integrand has only finitely many discontinuities. The tail diameter of A_{N_k} is at most 2^{1-h} for every h, hence zero.

Walsh's norm theorem [6, Theorem 1.1] gives $A_N \to L$ in L^2, using its cyclic-group specialization. Each rational skeleton's almost-everywhere limit equals L, by an almost-everywhere convergent subsequence of its L^2 convergence. For $c_i = 1 + i/s$, $0 \le i \le s$, and $2^k \le N < 2^{k+1}$, choose a lower grid index $n = \lfloor c_i 2^k \rfloor \le N$ with $N - n \le 2^k/s + 1$. Bounded summands give $|A_N - A_n| \le 2(N-n)/N \le 2/s + 2^{1-k}$. First take $k \to \infty$ for this finite grid, then $s \to \infty$. This proves the assertion on the countable conull set. Undo (1.3) and the normalization. □

If a norm theorem is stated only for standard probability spaces, the name map $x \mapsto (f_j(T^n x))_{j,n}$ pushes μ to a shift-invariant probability measure on a countable product of compact disks, and reproduces all the averages. It gives a standard Borel factor and a pullback proof; it is not a disintegration of the original space. The conditional conclusion is stronger than the mixing headline. That fact increases the importance of independently verifying L_b; it is not an unconditional solution of the general pointwise problem.

6.2 Weak mixing identifies the product, without a rate

Proposition 6.2 (Classical signed-slope norm identification)**.** *If T is weakly mixing, any finite collection of distinct nonzero integer slopes a_j and bounded inputs f_j satisfies $N^{-1}\sum_{n=1}^{N}\prod_j U^{a_j n}f_j \to \prod_j \int f_j\,d\mu$ in L^2. Consequently, under L_b, the M294 pointwise product limit holds with weak mixing in place of mixing.*

Proof. Normalize the inputs. The standard correlation characterization of weak mixing gives $H^{-1}\sum_{h=1}^{H}|\langle U^{ah}f, f\rangle| \to 0$ when $\int f = 0$ and $a \neq 0$. Nonzero powers preserve weak mixing; the signed assertion also follows by symmetry of correlations. Induct on the number of slots. For centered input in at least one slot, let $u_n = \prod_j U^{a_j n}f_j$ and $q_{j,h} = (U^{a_j h}f_j)\overline{f_j}$. For fixed h, invariance and induction at slopes $a_j - a_1$, $j \geq 2$, give

$$\frac{1}{N}\sum_{n=1}^{N-h}\langle u_{n+h}, u_n\rangle \longrightarrow \gamma_h = \prod_j \int q_{j,h}\,d\mu.$$

For a centered slot, $|\gamma_h|$ is bounded by its absolute correlation. Hence its Cesaro means tend to zero. The finite Hilbert van der Corput inequality, valid for $1 \leq H \leq N$, is

$$\begin{aligned}\|N^{-1}\textstyle\sum_n u_n\|^2 \leq \tfrac{N+H-1}{N^2H^2}\Big[H\sum_n\|u_n\|^2 \\ + 2\sum_{h=1}^{H-1}(H-h)\Re\sum_{n=1}^{N-h}\langle u_{n+h}, u_n\rangle\Big].\end{aligned} \tag{6.3}$$

It follows by zero extension, writing $H\sum_n u_n$ as the sum of $N+H-1$ sliding windows of length H, applying Cauchy–Schwarz, and expanding the squared window norms. Let $N \to \infty$ with H fixed to obtain an upper bound $H^{-1} + 2H^{-1}\sum_{h<H}|\gamma_h|$; then let $H \to \infty$. Expand general inputs into means and centered parts. The one-slot base is the mean ergodic theorem for the ergodic power T^{a_1}. This proves norm identification, a classical result credited to the weak-mixing/PET theory, not to Ram notation. Theorem 6.1 then proves the conditional pointwise assertion. □

There is no selection of $H = H(N)$ with a prescribed rate. Mixing alone does not provide one here. For signed smooth weights the source also uses Abel summation on positive and negative partial sums separately; both follow from the same norm theorem with slopes a_j and $-a_j$.

6.3 Affine shifts and periodic weights

Corollary 6.3 (Conditional bounded affine-periodic extension)**.** *Under L_b, for any fixed $r_j \in \mathbb{Z}$, any bounded periodic complex sequence w_n of period $q \geq 1$, and every bounded triple,*

$$\frac{1}{N}\sum_{n=1}^{N} w_n \prod_{j=1}^{3} f_j(T^{\lambda_j n + r_j}x)$$

converges almost everywhere. If T is weakly mixing its limit is $(q^{-1}\sum_{r=1}^{q} w_r)\prod_j \int f_j\,d\mu$.

Proof. Split the sum by $n = q\ell + r$, $1 \leq r \leq q$, with $\ell \geq 0$. In residue r use T^q and inputs $f_j \circ T^{\lambda_j r + r_j}$. The slopes remain λ_j and therefore have the same primitive b and local premise. The difference between starting at $\ell = 0$ and $\ell = 1$ is $O(1/N)$ for bounded summands. Each residue has asymptotic weight $1/q$, and a finite intersection of convergence sets suffices. Under weak mixing, T^q is weakly mixing and shifted inputs have the same integrals. Apply Proposition 6.2. □

This is a finite residue-class consequence, not an assertion for arbitrary bounded weights, primes, or point-dependent periods.

7 Stability and an unconditional model class

7.1 Negative-time perturbations and deterministic errors

Proposition 7.1 (Signed finite-horizon budget)**.** *Let U, V be Koopman unitaries on the same probability space, and let $\|U - V\|_{H\to H} \le \epsilon$. For one-bounded inputs,*

$$\|A_N^{\lambda,U}(\mathbf{f}) - A_N^{\lambda,V}(\mathbf{f})\|_2 \le \frac{N+1}{2}\left(\sum_j |\lambda_j|\right)\epsilon. \tag{7.1}$$

More generally, one-bounded decoded orbit fields with errors $\epsilon_{j,n}$ in L^2 give error at most $N^{-1}\sum_{n=1}^N \sum_j \epsilon_{j,n}$ in the average.

Proof. For negative powers, $U^{-1} - V^{-1} = U^{-1}(V - U)V^{-1}$, so the same operator norm bound holds. Telescoping powers gives $\|U^m - V^m\| \le |m|\epsilon$ for every integer m. Use (3.4) with $m = \lambda_j n$ and $N^{-1}\sum_{n=1}^N n = (N+1)/2$. The more general statement is exactly the bounded-bank product inequality. Radial clipping to the unit disk does not increase distance to a one-bounded true field, so bounded decoding can be enforced before using this bound. □

The premise on $U - V$ is strong. Generic small geometric changes of a transformation need not be small in this operator norm. An approximate linear model not known to be Koopman must separately control inverse iterates, bounded multiplication, and amplitude growth. These are finite-horizon results, with constants growing with N.

For finite schedules $Z_k, \widetilde{Z}_k$ with $\|Z_k - \widetilde{Z}_k\|_2 \le \epsilon_k$, the difference of their integral oscillation sums at fixed blocks is at most

$$\sum_i \left[(\sum_{k=k_i}^{k_{i+1}} \epsilon_k^2)^{1/2} + \epsilon_{k_i} \right]. \tag{7.2}$$

The maximum difference is bounded by the maximum of increment errors; bound that maximum by an ℓ^2 norm and integrate by Cauchy–Schwarz. This is the finite perturbation mechanism checked in M295, applicable with the signed orbit errors above. A finite set of such bounds does not meet the all-horizon premise of Lemma 5.2.

Proposition 7.2 (Bounded approximation in measure)**.** *For one-bounded triples $\mathbf{f}, \mathbf{h}$, any invertible T, and any nonzero signed slopes, put $e_j = \|f_j - h_j\|_1$. Then, for $t > 0$,*

$$\mu\{\sup_N |A_N^\lambda(\mathbf{f}) - A_N^\lambda(\mathbf{h})| > t\} \le \min\left\{1, \frac{(\sum_j \sqrt{e_j})^2}{t}, \frac{9}{t^2}\sum_j \|f_j - h_j\|_2^2\right\}. \tag{7.3}$$

Proof. Product telescoping bounds the maximal difference by $\sum_j M_{T^{\lambda_j}}|f_j - h_j|$, where $M_R a = \sup_N N^{-1}\sum_{n=1}^N a(R^n x)$ for $a \ge 0$. The positive ergodic maximal inequality has weak $(1,1)$ constant one, also for negative powers since those are probability preserving. Choose thresholds $t_j > 0$ summing to t. The union bound costs $\sum_j e_j/t_j$, minimized by $t_j = t\sqrt{e_j}/\sum_i \sqrt{e_i}$; zero terms can be omitted. For the third bound use Cauchy–Schwarz inside each average and thresholds $t/3$ for the resulting square roots, then the same positive maximal inequality on $|f_j - h_j|^2$. □

These inherited maximal principles give closure of any *actually dense* bounded convergence class. A proper characteristic-factor subspace is closed, not dense; norm characteristicity alone does not close the pointwise problem.

7.2 A finite-window Bernoulli example, proved directly

Proposition 7.3 (Cylinder inputs)**.** *On a two-sided iid product probability space with shift T, suppose $|f_j| \le 1$ and each f_j depends only on coordinates in $[-r, r]$, $r \in \mathbb{Z}_{\ge 0}$. For distinct nonzero signed slopes λ_j, $A_N^{\lambda} \to \prod_j \mathbb{E} f_j$ almost surely, without L_b.*

Proof. Let $Y_n = \prod_j f_j(T^{\lambda_j n} x)$ and $D_n = \bigcup_j ([\lambda_j n - r, \lambda_j n + r] \cap \mathbb{Z})$. For $n > 2r/\min_{i \ne j} |\lambda_i - \lambda_j|$, the three coordinate windows within Y_n are disjoint, so $\mathbb{E} Y_n = p := \prod_j \mathbb{E} f_j$. For fixed n, an overlap $D_n \cap D_m \ne \varnothing$ implies $|\lambda_i n - \lambda_j m| \le 2r$ for some i, j. For each pair at most $\lfloor 4r/|\lambda_j| \rfloor + 1 \le 4r + 1$ positive integers m satisfy it. Thus each n has at most $9(4r+1)$ possible dependent partners, including itself. Disjoint coordinate sets give independence. For the centered variables $\widetilde{Y}_n = Y_n - \mathbb{E} Y_n$, $\mathbb{E}|\widetilde{Y}_n|^2 \le 1$, so covariance has modulus at most one. Consequently

$$\mathbb{E}\left| N^{-1} \sum_{n=1}^{N} \widetilde{Y}_n \right|^2 \le \frac{9(4r+1)}{N}. \tag{7.4}$$

Chebyshev and Borel–Cantelli along $N = k^2$ give convergence to zero there, simultaneously at all rational positive error thresholds. The bounded-average gap estimate between k^2 and $(k+1)^2$ fills all integers. The finitely many initial expectations different from p have vanishing Cesaro contribution. □

This is a classical finite-dependence argument applied explicitly to the source's signed slopes, not a proof for arbitrary mixing systems. The sequence Y_n is not assumed stationary. For $r = 0$ and slopes $(-1, 2, 3)$, $Y_1 = X_{-1} X_2 X_3$ and $Y_2 = X_{-2} X_4 X_6$ are independent for fair $\{0, 1\}$ coordinates; Y_2 and $Y_3 = X_{-3} X_6 X_9$ share one coordinate. Their product expectations are $1/64$ and $1/32$, respectively, although each individual expectation is $1/8$. This directly refutes stationarity-based covariance shortcuts.

8 Tests, benefits, limitations, and future work

8.1 Concrete checks and counterexamples

The dedicated exact-arithmetic checker tests integer reconstruction, Vandermonde cancellation and quadratic identities across mixed and nonunimodular triples; square-variable moment formulas; cubic reproduction and the quartic Richardson error; finite cyclic orbit products and perturbation bounds; deterministic prefix variation; finite van der Corput inequalities; and Bernoulli overlap counts. Its JSON contains actual case and assertion counts, not extrapolated coverage: the author run passed 49,607 exact assertions across 503 primitive triples. These checks complement the displayed proofs. They do not establish L_b or certify every adaptive source construction.

For $(0, -6, 10, 15)$, $d_b = 21$ and a valid choice is $e = (1, -1, 1, -1)$. The checker also computes reconstruction vectors by extended gcd and verifies both equalities before use. The geometric formulas depend only on d_b once reconstruction is secured. For example at $\rho = 1/100$, $A_\rho = 1/52500$ and $\mathbb{E}\eta^2 = 1/6615000$.

On a circle rotation $T(x) = x + \alpha$, take slopes $(-1, 2, 3)$ and characters with frequencies $(2, 1, 0)$. Their product along the orbits is identically $e(3x)$ for every n because $-2 + 2 = 0$. The limit is not the product of means (which is zero). This is a convergent nonmixing example, demonstrating the necessity of a limit-identification hypothesis in Theorem 6.1. On a finite cyclic system the same resonance is exact modulo the period. Such systems are not nontrivial mixing models.

The bounded scalar sequence $\sin\log(k+2)$ has square-summable adjacent increments, since their modulus is at most $1/(k+2)$, but does not converge. This is a counterexample to inferring convergence from adjacent square energy, not a counterexample to the source ergodic claim. A fixed finite orbit likewise cannot determine an infinite tail diameter without a uniform transfer argument.

8.2 What is improved, and what is inherited

The realized additions are the exact arbitrary-slope polygon moment formula, the two-radius fourth-order summed error and its explicit $17/3$ constant penalty, and signed-time propagation of reconstruction errors with $\sum_j |\lambda_j|$. The cylinder-input example is a fully proved model-class check. The conditional general-system and affine-periodic consequences expose the actual scope of the local premise. Classical interpolation, Gram geometry, maximal inequalities, van der Corput, norm convergence, and Richardson extrapolation are attributed tools. M295 already contains the $(1, 2, 3)$ second-order planting improvement, linearization factor 2, robust oscillation criterion, and bounded approximation estimates; they are not counted again as M294 discoveries.

Advantages are concrete: negative iterates have a declared operator domain; no false unimodular-pair assumption enters planting; the discretization error has auditable constants and powers; all block endpoints remain deterministic; and conditional consequences do not depend on an invented mixing rate. The operator bank can expose finite-model errors and resonances without conflating coefficient energy with physical energy.

The costs are also concrete. Fourth-order extrapolation uses two plantings, signed coefficients, higher profile derivatives, and a larger leading bound. Small planting radii make A_ρ^{-1} large. Moment correction pushes support outward, enlarging orbit padding and profile constants. Finite banks scale with horizon and input complexity. No claimed accuracy gain proves reduced runtime. Most importantly, none of these improvements resolves the uniform localized multilinear estimate or its imported row and counting dependencies.

Future work should compare the two explicit planting bounds using actual profile derivatives and support costs; verify the source's dimension-polynomial rational-shift reserves and scheduled Bessel premises independently; test finite data with negative and highly nonunimodular slopes; and develop rigorously controlled decoding errors on a genuine approximation class. An effective convergence rate would require additional quantitative hypotheses and a new argument. These are proposed tasks, not completed contributions.

9 Read scope and provenance

The root is `C:/Users/mural/OneDrive/Documents/OpenAI-722-math`. All eight TeX files in the following path (concatenating the three lines)

```
_sources/openai-math/preprints/
Triple-ergodic-averages-with-distinct-integer-slopes-
October-4-2026/build
```

were read: `main.tex`, `references.tex`, and the six sections `introduction.tex`, `local.tex`, `transfer.tex`, `analytic.tex`, `arithmetic.tex`, `completion.tex`. The proof map names the actual assertions used. The companion H and V proofs were not re-audited in their entirety in this author assignment; source imports remain stated dependencies.

The actual private `res-math/TR-item-168-astra.tex` under the `res-rt` workspace was read in full, including its corrected deterministic oscillation proof, fine-skeleton argument, bounded-input closure, and finite-horizon budgets. The delivered `M295-Triple-Ergodic-Averages/`

`TR-M295-Ram-Triple.tex` was read in the actual relevant passages: lines 369–582 (oscillation, generated factor, grids, weak mixing) and 1107–1410 (kernel, planting, linearization, physical Gram and perturbation arguments), with additional contextual passages encountered. Its references and reported test totals are not adopted as fresh evidence for M294. No whole-corpus read is claimed.

The author also read the entire completed independent source audit (concatenating these two path lines):

```
_audit/
M294-independent-source-review.txt
```

The review read all eight target files and selected, explicitly delimited H/V passages; its 16 byte-matched source identities do not assert complete reading of every companion file. Its separate 33,174 exact bounded checks, together with the listed fourth-moment counts, support the particular algebra and parameter checks stated there, not the whole analytic claim. Unfinished certification includes the full adaptive base and projection construction, rank closure and localization, mode coherence and top counting, complete side stacks, scheduled-row applications, and external inverse, extrapolation, and geometry-of-numbers inputs. Its pass concerns conditional transfer. There is neither an unconditional pass nor a source counterexample. In particular, the review's counterexample to an unused fixed-phase multiplier bound is not a refutation of the source's averaged fourth-moment bound. A byte-preserving evidence snapshot records the review, its read scope, source manifest, and checks.

Dedicated evidence records store SHA-256 identities, public URL check scope and failures, exact local tests, native compiler outcome, physical export logs, and page inspection. Hash identity is not proof verification. Original source files and shared project records are unchanged. The mathematical conclusion remains conditional on L_b, irrespective of document compilation, finite tests, or the bounded independent review.

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